NATIONAL TECHNICAL UNIVERSITY OF ATHENS

School of Electrical and Computer Engineering

Machine Learning-Based Stress Detection

During COVID-19 Diagnostic Testing

Extended English Summary of Diploma Thesis

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Document status. This document is an author-prepared, selective English account of an original diploma thesis submitted in Greek. It is *not* a certified translation or a replacement for the original thesis. Figures reproduced below are taken from the original document; the comparative score chart is newly prepared from values reported in Chapter 7.

Abstract

This diploma thesis investigated the use of galvanic skin response (GSR) measurements and machine-learning classifiers to distinguish observations associated with elevated stress from those associated with a more relaxed state during COVID-19 rapid diagnostic testing. The experimental workflow comprised sensor-data preprocessing, the extraction of signal-derived features, an initial clustering-assisted classification experiment, and a broader comparative study of support vector machines (SVMs). For the principal model comparison, GSR observations from the Young Adult's Affective Data (YAAD) dataset were used for training, while COVID-19 testing measurements and selected GSR observations from an esports dataset were used for testing, as described in the thesis. Linear, radial basis function (RBF), polynomial, sigmoid, and Nu-SVM configurations were compared. In the reported experiments, the best accuracy was **70%** and the best F1-score was **75%**; these results were obtained in *different* model configurations. The work illustrates practical machine-learning experimentation on physiological data while showing the sensitivity of classification performance to kernel and parameter choices. **Keywords:** Machine learning; physiological signals; galvanic skin response; electrodermal activity; feature extraction; support vector machines; stress classification.

Original full thesis (Greek): https://artemis.ece.ntua.gr/items/c339a9e9-8280-43f7-b99f-23e70102e48d

1 Research problem and physiological signal

The research question was whether a computational pipeline based on physiological skin-conductance measurements could help classify a subject as "stressed" or "relaxed" in the context of a COVID-19 diagnostic test. GSR, also described as electrodermal activity, measures changes in skin conductance arising partly from sympathetic nervous system activity. GSR indicates physiological arousal, however it does not, on its own, provide a direct or definitive diagnosis of psychological stress.

The original study describes sensor acquisition using an Arduino-associated arrangement and discusses how readings can be converted from sensor units into electrical quantities and eventually conductance measurements, expressed in microsiemens (μS). The experimental analysis focused on a compact numerical representation rather than feeding entire raw waveforms to the SVM classifiers.

2 Data preparation and feature engineering

2.1 Data inspection and transformation

The spreadsheet measurements were inspected for missing or inconsistent entries. Electrical quantities were derived from the sensor readings, followed by feature extraction from the GSR series. The thesis describes normalization to place the selected numerical features on comparable scales.

Feature or quantity	**Role in the analysis**
Mean GSR	Average conductance level over the considered observations.
Peak amplitude	Captures a maximum response-related signal magnitude.
Rise time	Time interval associated with the rise of the response.
Latency	Delay between the reference event and the beginning of the response.
Peak amplitude − mean	Derived contrast between the signal's peak and its mean; emphasized in the experimental comparison.

Table 1: GSR-derived quantities discussed in Chapter 5. The primary two-dimensional model comparison used mean GSR and peak-amplitude-minus-mean.

	Mean	Peak-Mean	Rise Time	Peak Amplitude (µS)	Latency (sec)
Test 1	1.56	0.06	33	0.14	5
Test 2	2.58	0.37	26	0.88	16
Test 3	2.28	0.71	8	1.06	6

Figure 1: Example of the prepared tabular feature representation. *Figure 12 of the submitted Greek-language thesis.*

The final feature subset for the main classification illustrations comprised **mean GSR** and **peak amplitude minus mean GSR**, enabling two-dimensional plots of decision regions. The original thesis also discusses additional GSR timing characteristics, but these should not be confused with the final two-feature inputs of every experiment.

3 Preliminary clustering-assisted experiment

The first approach considered a **small subset of nine GSR observations** collected during COVID-19 testing. Because these observations initially lacked stress labels, the experiment used **K-Means clustering** (implemented with Python's `scikit-learn`) to group data into two categories. Those cluster assignments were then used as labels in an SVM illustration. The original experiment used an 80% training / 20% test split (two test observations).

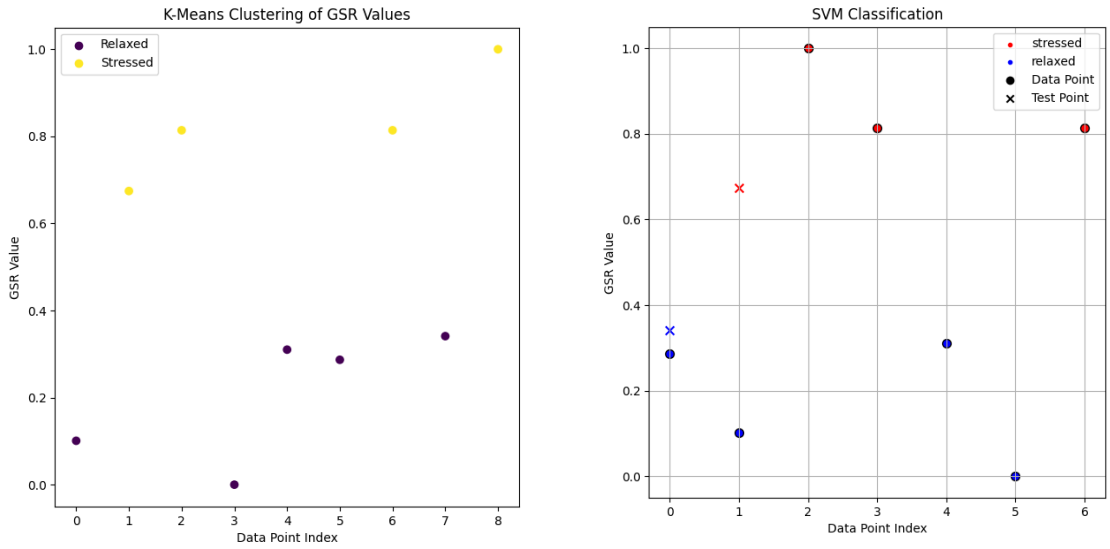


Figure 2: Preliminary clustering (left, *Figure 13 of the submitted Greek-language thesis.*) and subsequent SVM classification (right, *Figure 14 of the submitted Greek-language thesis.*). Images reproduced directly from the source thesis.

The thesis reports **100% accuracy** in this preliminary demonstration. Crucially, that number describes only the very small and easily separable dataset of nine observations. Furthermore, the provisional labels were generated from the same measurements through clustering rather than independently established clinical ground truth. For these reasons, the result should be understood as a proof-of-concept illustration of a classification pipeline, not evidence of 100% stress-detection performance in practical use.

3.1 Transition to a broader evaluation

The initial experiment motivated a more varied comparison of SVM decision functions. The second approach, detailed in the following sections, used an external training set and separate test measurements, evaluated a broader family of kernels, and provided both overall accuracy and F1-scores for the candidate configurations.

Methodological distinction. The 100% result from the preliminary nine-observation illustration is *not* part of the comparative performance table in Section 6. The latter table reproduces the larger SVM/Nu-SVM evaluation from Chapter 7.

4 Main SVM experiments and datasets

4.1 Experimental sources and feature space

The main classification experiments trained SVM models using GSR measurements from **YAAD (Young Adult's Affective Data)**, which the thesis describes as a dataset with ECG and GSR measurements from 25 participants. Testing used GSR measurements collected during COVID-19 diagnostic tests together with selected GSR data from an external **esports** dataset discussed in Chapter 6. Other modalities from these datasets were not used in the described classification experiments. The experiments plotted the two numerical features *mean GSR* and *peak amplitude minus mean GSR*.

4.2 Kernel-based classification

A linear SVM separates observations with a straight decision boundary in the two-dimensional feature representation. An RBF SVM can form curved decision boundaries; the specific RBF example reproduced below uses $\gamma = 2.0$. The thesis also explored polynomial decision boundaries with degrees 5 and 10 and a sigmoid-kernel SVM.

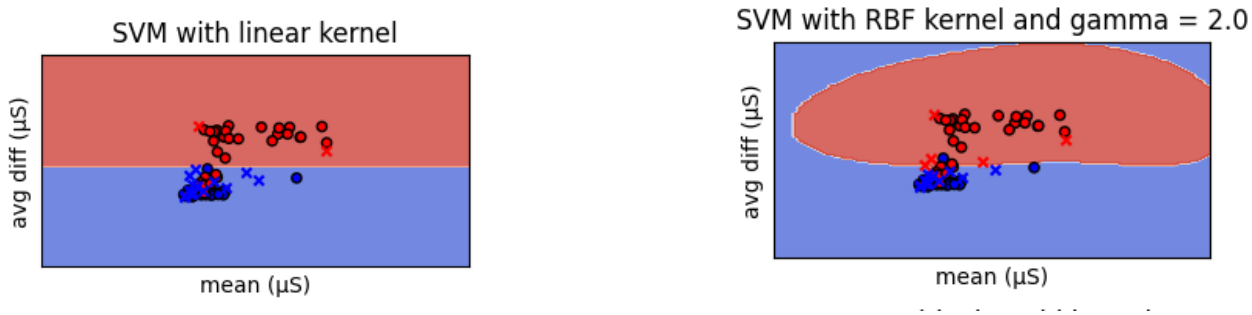


Figure 3: Linear-kernel SVM (left, *Figure 15 of the submitted Greek-language thesis.*) and RBF-kernel SVM with $\gamma = 2.0$ (right, *Figure 17 of the submitted Greek-language thesis.*).

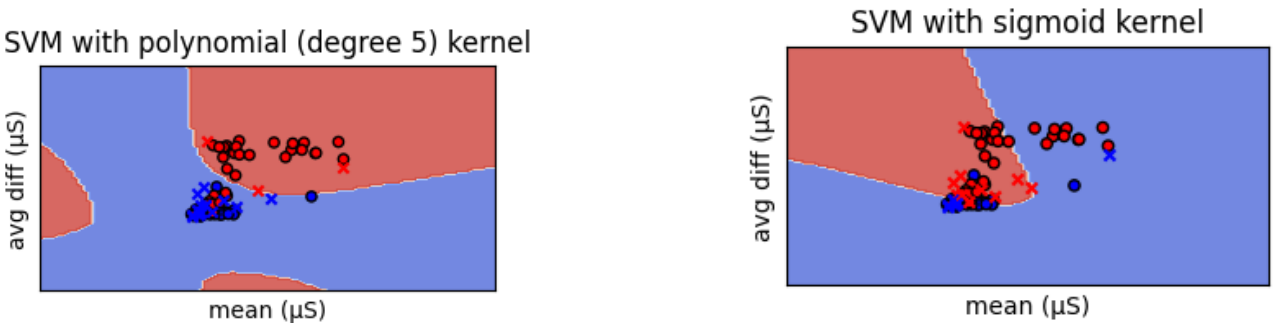


Figure 4: Polynomial-kernel SVM, degree 5 (left, *Figure 18 of the submitted Greek-language thesis.*) and sigmoid-kernel SVM (right, *Figure 20 of the submitted Greek-language thesis.*).

These figures visualize differences in model geometry. A more elaborate boundary does not automatically imply better performance on unseen test samples. The following evaluation consequently compares numerical scores rather than judging models only by the appearance of their decision boundaries.

5 Nu-SVM and parameter sensitivity

In addition to standard SVMs, the thesis investigated **Nu-SVM** classifiers. Nu-SVM changes the parameterization of the support-vector optimization problem; the experimental work varies ν to examine how classification regions and reported performance change under the tested conditions.

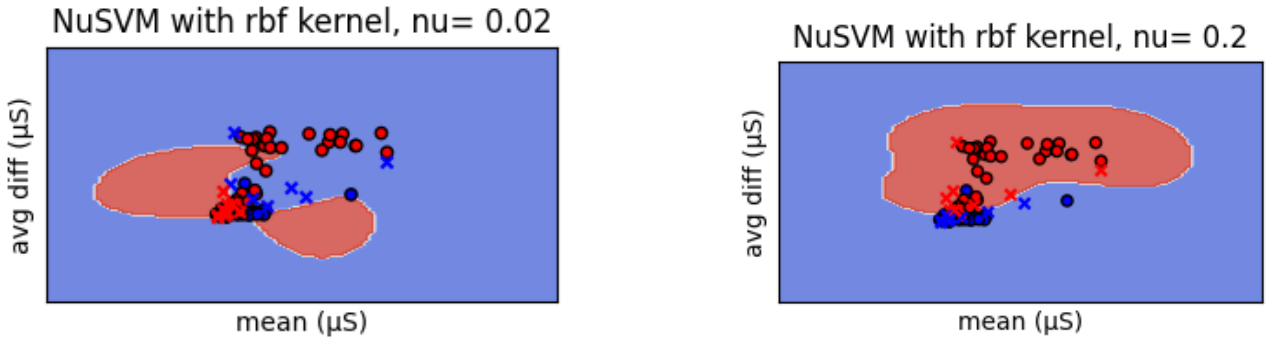


Figure 5: Nu-SVM with RBF kernel: $\nu = 0.02$ (left, *Figure 21 of the submitted Greek-language thesis.*) and $\nu = 0.2$ (right, *Figure 22 of the submitted Greek-language thesis.*).

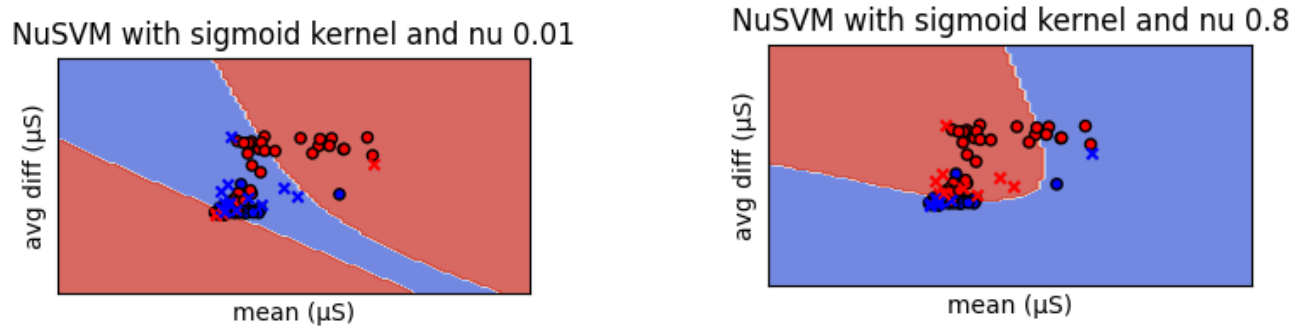


Figure 6: Nu-SVM with sigmoid kernel: $\nu = 0.01$ (left, *Figure 23 of the submitted Greek-language thesis.*) and $\nu = 0.8$ (right, *Figure 24 of the submitted Greek-language thesis.*).

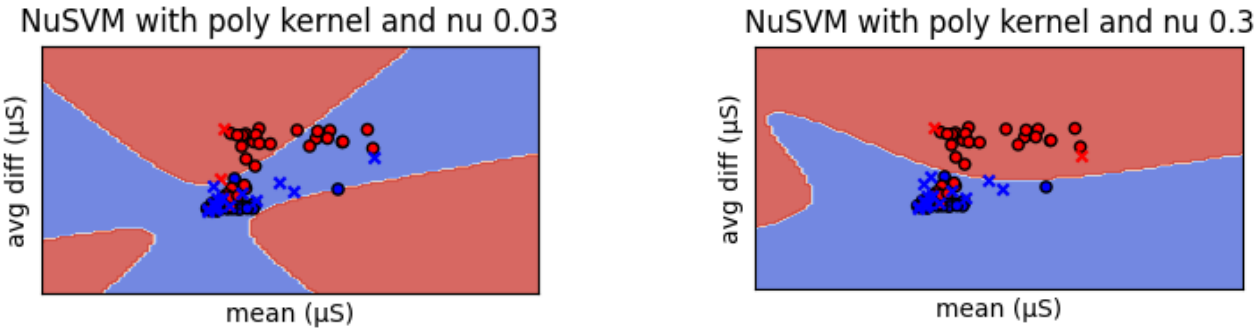


Figure 7: Nu-SVM with polynomial kernel: $\nu = 0.03$ (left, *Figure 25 of the submitted Greek-language thesis.*) and $\nu = 0.3$ (right, *Figure 26 of the submitted Greek-language thesis.*).

The tested RBF values show a relevant trade-off: the $\nu = 0.02$ setting achieved the **highest F1-score** in the comparison, whereas $\nu = 0.2$ reached the joint-highest **accuracy**. The sigmoid Nu-SVM was also strongly sensitive to the tested value of ν. Polynomial Nu-SVM variants were included in the numerical comparison; the $\nu = 0.03$ example had especially low scores, while raising ν to 0.3 improved them only modestly.

6 Quantitative comparison of models

6.1 Reported performance measures

The original thesis assesses its main experiments using **accuracy** (fraction of correctly classified test cases) and **F1-score** (harmonic mean of precision and recall). These are distinct quantities. Individual precision and recall values for every model are not tabulated below because the original Chapter 7 reports the pair of scores shown, not a complete precision/recall table.

Classifier / configuration	Accuracy	F1-score
Linear-kernel SVM	40%	25%
LinearSVM (specialized linear implementation)	50%	44.44%
RBF-kernel SVM	50%	44.44%
Polynomial-kernel SVM, degree 5	40%	25%
Polynomial-kernel SVM, degree 10	40%	25%
Sigmoid-kernel SVM	**70%**	72.73%
Nu-SVM / RBF, $\nu = 0.02$	60%	**75%**
Nu-SVM / RBF, $\nu = 0.2$	**70%**	72.73%
Nu-SVM / sigmoid, $\nu = 0.01$	30%	36.36%
Nu-SVM / sigmoid, $\nu = 0.8$	60%	60%
Nu-SVM / polynomial, $\nu = 0.03$	30%	0%
Nu-SVM / polynomial, $\nu = 0.3$	40%	25%

Table 2: All configurations described in Section 7.1 of the original thesis. Bold values mark the highest reported accuracy or F1-score; the two maxima do not occur in the same setting.

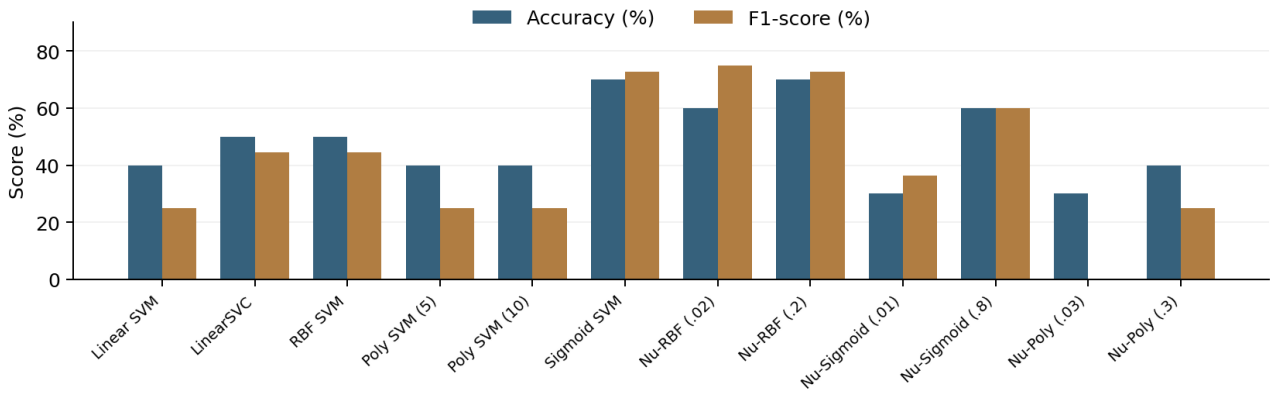


Figure 8: New comparison chart reconstructed from the original Chapter 7 score values (not a figure from the original thesis).

7 Interpretation, limitations and future work

7.1 Principal experimental findings

The strongest *accuracy* was **70%**, obtained both by the standard sigmoid-kernel SVM and by Nu-SVM with an RBF kernel and $\nu = 0.2$. The best *F1-score* was **75%**, achieved by the RBF Nu-SVM with $\nu = 0.02$; its corresponding accuracy was 60%. These differences demonstrate why the ranking of classifiers depends on the evaluation metric. Conversely, the polynomial Nu-SVM at $\nu = 0.03$ returned 30% accuracy and a zero F1-score, highlighting the presence of less successful parameter settings in the same study.

7.2 Important limitations

- **Small preliminary dataset.** The initial nine-measurement demonstration had only two test observations and used cluster-based labels. Its 100% score cannot establish generalization to an independent population.
- **GSR is an indirect measure.** Skin-conductance changes are informative for physiological arousal but are not uniquely attributable to psychological stress.
- **Dataset and setting differences.** The main experiments drew training and test measurements from different collections and contexts; sensor, population and acquisition differences can limit generalization.
- **Model-selection variability.** Kernel type and Nu-SVM settings changed both decision regions and the observed scores; the outcomes should not be extrapolated beyond the reported evaluation.
- **No clinical validation is claimed.** The thesis evaluates classification experiments, not a validated diagnostic device or deployable clinical system.

7.3 Directions discussed in the thesis

Chapter 8 identifies possible future work combining GSR with additional signals such as heart-rate variability and EEG, investigating deep-learning or ensemble-based approaches, and designing non-invasive systems suitable for more continuous monitoring. These proposals are *future research directions* and are not presented as techniques implemented or validated in the 2024 experiments.

8 Relevance to a Master's in Data Science and Artificial Intelligence

This thesis documents applied experience in sensor-derived data preparation, numerical feature engineering, clustering, supervised SVM classification, experiment design, comparison of hyperparameters and kernels, and performance evaluation. The study is a record of *research and programming experience*; it is not presented as formal completion of a standalone undergraduate Machine Learning course.

Original academic source

Bitsikas, G. (2024). *Assessment of Stress During COVID-19 Diagnostic Testing Using Machine Learning* [Diploma thesis, School of Electrical and Computer Engineering, National Technical University of Athens; original in Greek]. https://artemis.ece.ntua.gr/items/c339a9e9-8280-43f7-b99f-23e70102e48d

Note: All methods, numbers and source figures referenced above are derived from the original submitted thesis. Descriptive caveats clarify the evidential limits of those results and do not represent additional experiments.